matrices and align

Paperflow Test Suite

November 2026

1 Linear Algebra, Matrices & Aligned Structures

Author: Applied Mathematics Laboratory

1.1 1. Matrix Inversion & Transformations

Given a square 2×2 real matrix $\mathbf{M} \in \mathbb{R}^{2\times 2}$:

$$\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{1}$$

The determinant is $\det(\mathbf{M}) = ad - bc$. Assuming $\det(\mathbf{M}) \neq 0$, the inverse is:

$$\mathbf{M}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{2}$$

For a general 3×3 matrix:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{3}$$

1.2 2. Aligned System of Equations

Maxwell's microscopic equations in differential form:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned} \tag{4}$$

Derivation of energy flux continuity via Poynting's theorem:

$$
\begin{aligned}
\mathbf{S} &= \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}) \\
u &= \frac{1}{2}\left(\varepsilon_0 \parallel \mathbf{E} \parallel^2 + \frac{1}{\mu_0} \parallel \mathbf{B} \parallel^2\right) \\
\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} &= -\mathbf{J} \cdot \mathbf{E}
\end{aligned} \tag{5}
$$

1.3 3. Piecewise Functions and Cases

The Heaviside step function and Dirac delta limit:

$$
\Theta(x) = \begin{cases} 0, \text{if } & x < 0 \\ \frac{1}{2}, \text{if } & x = 0 \\ 1, \text{if } & x > 0 \end{cases} \tag{6}
$$

The signum function:

$$
\operatorname{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases} \tag{7}
$$